

Differentially Private Anonymous Bandits for Multi-User Systems

Mohammad Reza Badri
mohammadreza.badri@epfl.ch
EPFL
Lausanne, Switzerland

Samira Hossein Ghorban
s.hosseinghorban@ipm.ir
Institute for Research in Fundamental Sciences
Tehran, Iran

Hossein Esfandiari
esfandiari@google.com
Google Research
London, United Kingdom

Alireza Rezaeimoghadam
alireza.moghadam@sharif.edu
Sharif University of Technology
Tehran, Iran

Abstract

In this work, we develop a unified *multi-user multi-armed bandit framework* that integrates *differential privacy (DP)* with *anonymous feedback* across three privacy models: (1) a *centralized DP* model, where a trusted server adds noise; (2) a *Local DP* model, where users randomize their own data; and (3) a novel *Semi-Central DP* model, in which a trusted *anonymity agent* aggregates and perturbs rewards before passing them to an untrusted server.

Our algorithms enable N users to collaborate in an ϵ -differentially private and C -anonymous manner, while maintaining low regret. We establish regret bounds of $\tilde{O}(NC\sqrt{KT\alpha}/\epsilon)$ under the Local DP model, $\tilde{O}(NC\sqrt{KT\alpha} + N(C\alpha)^{3/4}T^{1/4}K/\epsilon)$ under the Central DP model, and $\tilde{O}(NC\sqrt{KT\alpha} + N\sqrt{CKT\alpha}/\epsilon)$ under the Semi-Central DP model, where T denotes the time horizon, K denotes the number of arms, and α represents a decomposition overhead factor.

Our results provide the first unified treatment of privacy and anonymity in multi-user multi-armed bandit learning, revealing the trade-offs between collaboration, privacy protection, and learning efficiency.

CCS Concepts

• **Theory of computation** → *Sequential decision making*; **Online learning algorithms**; • **Security and privacy** → Data anonymization and sanitization; **Privacy protections**.

Keywords

Multi-Armed Bandits, Differential Privacy, Anonymity

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1 Introduction

In many modern technologies, such as recommendation systems or online markets, we need to decide when to try something new

(explore) and when to stick with what already works well (exploit). The goal is to earn as many rewards as possible over time. This fundamental challenge, formalized as the classical bandit problem, becomes significantly more complex when multiple users interact with the system simultaneously, each requiring personalized decisions while maintaining strict privacy guarantees [3, 28, 34, 37]. Online healthcare platforms often aim to recommend effective treatments by learning from patient outcomes. To do this well, they must identify patterns across different patient profiles while ensuring that no individual's medical details are revealed [10, 27]. Similarly, recommendation systems personalize content for millions of users, but must do so without exposing anyone's private preferences or behaviors. These examples illustrate a central challenge: how to design learning systems that adapt to users' needs without compromising their privacy.

The intersection of privacy concerns and sequential decision-making has gained critical importance as personalized systems handle increasingly sensitive user data [4, 40]. Traditional bandit algorithms assume complete observability of individual rewards, creating an inherent tension with privacy requirements [36]. This problem becomes more serious when the collaboration of the users is involved [39, 41]. Privacy regulations, including the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), have made addressing these concerns not only ethically important but legally mandatory for many applications [20].

Multi-user bandit systems have come a long way, starting with individual learning models, moving into collaborative strategies. For the first time, single-user bandits were studied by the foundational work of Robbins [30]. Later, Lai and Robbins (1985) established asymptotic optimality and regret lower bounds in the single-user setting [23]. Auer et al. (2002) introduced UCB algorithms with finite-time bounds that are still central to the field today [3].

The extension of bandit algorithms to multi-user settings has led to significant developments in collaborative learning. Sankaranarayanan et al. introduce one of the earliest models of social learning in multi-agent bandits [32]. Building on this foundation, recent contributions include a variant of the UCB algorithm that achieves logarithmic regret under client sampling [40]. Together, these advances illustrate the field's progression from single-agent learning to more collaborative paradigms that also account for privacy and system-level robustness.



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1.1 Anonymous Learning

Our work builds directly upon the recent framework on anonymous bandits for multi-user systems [16]. Their approach introduced k -anonymity in online learning settings, ensuring that observations are aggregate rewards of at least k users, which prevents singling out individuals while allowing group-based learning.

The concept of k -anonymity itself was originally formalized by Samarati and Sweeney [31] and further developed in Sweeney's comprehensive model with quasi-identifiers [34]. Important extensions include ℓ -diversity [26] and t -closeness [24], which address homogeneity and background knowledge attacks that can compromise simple k -anonymity approaches. However, k -anonymity represents a syntactic privacy approach with known limitations, particularly against adversaries with auxiliary information.

A critical distinction exists between static anonymization techniques (traditional k -anonymity) and the dynamic anonymous learning paradigm introduced by [16], which aggregates at least C user rewards per observation in an adaptive and real-time manner while learning user similarities without individual observations.

1.2 Differential Privacy in Bandit Settings

Differential privacy, introduced by [14], provides mathematically rigorous guarantees that individual data points cannot be distinguished through algorithmic outputs [14]. These foundational ideas were further formalized and expanded in the influential monograph of [15]. Applying these ideas to sequential decision-making introduces unique challenges, as algorithms must safeguard privacy throughout ongoing interactions while continuing to learn effectively.

Recent progress has led to major advances in this area. Wang and Zhu established tight lower bounds, showing that under certain conditions ($\delta > 0$), privacy is achieved with almost optimal long-term cost in multi-armed bandits [40]. More recently, in [5], key gaps between upper and lower regret bounds are resolved in Bernoulli bandits under global differential privacy.

Hanna et al. achieve differential privacy in stochastic linear bandits in central and local primary differential privacy models [19]. In the *Local DP* model, an untrusted server receives privatized rewards from clients directly. In the *Central DP* model, a trusted server applies a DP mechanism to aggregated rewards before using them. In the Central DP model, they achieve a regret of $O(\sqrt{T \log T} + \frac{\log^2 T}{\epsilon})$, matching existing lower bounds up to logarithmic factors. This is particularly important since it shows that for typical privacy parameters ($\epsilon \approx 1$), the dominant term remains $O(\sqrt{T \log T})$, which matches non-private algorithms. For the local privacy model, they achieve $O(\frac{\sqrt{T \log(T)}}{\epsilon_0})$ regret. Their approach leverages carefully selected core sets of actions to minimize the overall amount of noise needed for privacy guarantees.

1.3 Embedding DP and Anonymity together

While both anonymity and differential privacy (DP) are powerful privacy-preserving tools, they offer distinct and complementary guarantees. Anonymity is intuitive and easy for users to understand, providing a tangible sense of security by ensuring individuals are

hidden within a group [16]. However, it can be vulnerable to sophisticated inference attacks from adversaries possessing auxiliary information.

In contrast, DP offers rigorous, mathematical guarantees against such attacks and can be integrated into bandit algorithms with minimal performance impact [4, 35, 40]. One potential challenge is that the complexity of its probabilistic guarantees can make them more difficult for non-experts to interpret, which may affect user trust.

By integrating these paradigms, we create a layered privacy framework that leverages the strengths of both. Anonymity serves as an understandable first line of defense, while DP provides a formal mathematical backstop. This combined approach yields a more comprehensive and trustworthy system where privacy is both user-friendly and mathematically robust.

1.4 Our Contributions

We study the problem of online learning in multi-user systems under strong privacy constraints, focusing on the integration of differential privacy with anonymous learning paradigms.

By carefully designing algorithms that aggregate information across users while adding calibrated noise, we achieve no-regret algorithms which preserve strong privacy guarantees. Specifically, we develop batched successive elimination algorithms that operate on anonymous groups of users, incorporating differential privacy through noise mechanisms that preserve the utility of aggregated feedback.

Our approach builds on two key theoretical foundations. First, [16] provide an analysis of preserving anonymity for MABs, establishing near-matching upper and lower regret bounds for their proposed U -batched assumption. We extend their framework by incorporating privacy constraints, exploring how their algorithm could be efficiently modified to preserve privacy. By combining differential privacy with anonymity, our algorithms offer stronger protection than either approach alone.

Our main contributions are:

- (1) **Unified Framework:** We present the first comprehensive framework integrating differential privacy with anonymous bandits, providing algorithms for multiple privacy models including Local DP, Central DP, and a novel anonymous DP setting where the aggregation server is trusted but the learning algorithm is not which we call Semi-Central DP.
- (2) **Anonymous DP Multi-Armed Bandits:** We develop ADP-BaSE (Anonymized Differentially Private Batched Successive Elimination), based on instances of the BaSE algorithm [18] for the local and Central DP models which have been investigated in the bandit literature without anonymity constraints [19]. Our method achieves the following regret bounds with high probability:
 - $\tilde{O}\left(\frac{NC\sqrt{KT}\alpha}{\epsilon}\right)$ in the Local DP model.
 - $\tilde{O}\left(NC\sqrt{KT}\alpha + \frac{N(C\alpha)^{3/4}T^{1/4}K}{\epsilon}\right)$ in the Central DP model.

Where, $T, N, K, C, \epsilon, \alpha$ denote the time horizon, the number of users, the number of arms, the anonymity parameter, the privacy parameter and the decomposition overhead factor respectively.

- (3) **Semi-Central DP Multi-Armed Bandits:** Inspired by the way local and Central DP models work with the presence of Anonymity, we introduce a new model of privacy which we call Semi-Central. In the Semi-Central DP model, the anonymity agent (which is defined in the presence of anonymity constraints) is trusted and applies a DP mechanism to aggregated rewards before sending them to the server. In this novel model, we are able to show that ADPBaSE achieves a regret of $\tilde{O}\left(NC\sqrt{KT\alpha} + \frac{N\sqrt{CKT\alpha}}{\epsilon}\right)$ with high probability. We show that this model can provide both anonymity and an almost local form of differential privacy while maintaining nearly the same level of regret. By adjusting the model parameters, it is also possible to achieve regret comparable to that of a setting without differential privacy. This demonstrates that both privacy guarantees can be achieved together with almost no additional cost.

1.5 Challenges

To address anonymity challenges, we require a feedback elicitation algorithm that can estimate each user's rewards. The feedback elicitation method proposed in [16] produces many dependent estimations across users, which prevents the model from satisfying differential privacy. We resolve this issue by introducing a new feedback elicitation algorithm that generates independent estimations for users. Furthermore, to ensure differential privacy in both the Semi-Central and Central DP models, we modify the algorithm to combine these estimations after adding Laplace noise.

To bound the regret across all models, it is necessary to control two types of noise: the standard stochastic noise in the feedback (combined with the anonymity penalty) and the Laplace noise (combined with either the anonymity or local privacy penalty, depending on the model). The main challenge arises from the concentration inequality of the average of Laplace noises, which yields two different bounds depending on the concentration parameter. In each model, the appropriate bound must be chosen based on the number of independent noise terms and first batch size in order to determine the correct accuracy parameter.

2 Models and preliminaries

Notations. We say a random variable X is σ^2 -sub-Gaussian if the mean-normalized variable $Y = X - \mathbb{E}[X]$ satisfies $\mathbb{E}[\exp(sY)] \leq \exp(\sigma^2 s^2/2)$ for all $s \in \mathbb{R}$. We write $[N]$ as shorthand for the set $\{1, 2, \dots, N\}$.

2.1 Stochastic Multi-Armed Bandits

We consider the classical *stochastic multi-armed bandit* (MAB) problem. The learner interacts with an environment over T rounds. In each round $t \in \{1, \dots, T\}$, the learner selects an arm $a_t \in \mathcal{A}$ from a finite set of K arms, $\mathcal{A} = \{1, \dots, K\}$. Upon selecting arm a_t , the learner receives a stochastic reward $r_t(a_t) \in [0, 1]$, drawn independently from an unknown distribution $\mathcal{D}_{a_t}$ with expectation $\mu_{a_t} := \mathbb{E}[r_t(a_t)]$.

The goal of the learner is to maximize the cumulative expected reward or, equivalently, minimize the *expected cumulative regret*:

$$\mathbb{E}[\mathcal{R}_T] = T\mu^* - \mathbb{E}\left[\sum_{t=1}^T r_t(a_t)\right],$$

where $\mu^* := \max_{a \in \mathcal{A}} \mu_a$ is the mean reward of the optimal arm.

2.2 Batched Stochastic Bandits

In the *batched bandit* setting, the learner is restricted to interacting with the environment in B batches. Let the timeline be partitioned into B disjoint batches $b_1, \dots, b_B$, where batch b contains D_b rounds, and $\sum_{b=1}^B D_b = T$. At the beginning of each batch b , the learner chooses a policy (e.g., a probability distribution over arms or actions) based on all past observations up to the end of batch $b-1$, and applies it throughout batch b .

The learner only updates its strategy at the end of each batch, rather than after each individual round. This constraint models scenarios where decisions must be made in parallel or under communication limitations.

We will use R_b to denote the regret incurred in batch b .

Batched bandits introduce a trade-off between statistical efficiency and adaptivity: fewer batches mean less frequent updates, which can lead to higher regret, but are often necessary in practical systems with limited interaction capability.

2.3 Anonymity in Bandits

We consider the *anonymous multi-user bandit* setting introduced by [16]. There are N users and an action set $\mathcal{A}$ of K arms (as in a standard stochastic bandit). Each round $t = 1, 2, \dots, T$, the learner (platform) selects an arm $a_{t,i} \in \mathcal{A}$ for each user $i \in [N]$. If user i plays arm a in round t , they receive a reward $r_{t,i}(a)$ drawn from an unknown distribution $\mathcal{D}_{i,a}$ with mean $\mu_{i,a}$. We assume these reward distributions are independent and the goal is to minimize the cumulative regret across all users.

The key distinction in this model is that feedback is *anonymous* and aggregated. The learner does not observe individual rewards $r_{t,i}$, but only certain aggregated feedback from groups of users. In each round, the algorithm may partition a subset of users into groups of size at least C , under the constraint that all users in one group are assigned the same arm. For each such group G , the learner observes only the total reward $\sum_{i \in G} r_{t,i}$ obtained by that group (while no feedback is received for any user not in a group). This C -anonymity constraint (inspired by k -anonymity) ensures that each recorded observation aggregates at least C users' outcomes, so individual user behavior is indistinguishable in the feedback. The grouping (anonymization) pattern can be chosen anew in each round, but at all times any feedback must come from a group of size at least C .

The expected cumulative regret for this problem is given by

$$\begin{aligned} \mathbb{E}[\mathcal{R}_T] &= \sum_{t=1}^T \left[\sum_{i=1}^N \mu_{i,a_t^*} - \sum_{i=1}^N \mu_{i,a_{t,i}} \right] \\ &= \sum_{t=1}^T \sum_{i=1}^N (\mu_{i,a_t^*} - \mu_{i,a_{t,i}}) \end{aligned}$$

where for each user i ,

$$a_i^* = \arg \max_{a \in \mathcal{A}} \mu_{i,a}$$

is the optimal arm for user i , and $\mu_{i,a}$ is the (unknown) mean reward of arm a for user i .

2.4 Differentially Private Bandits

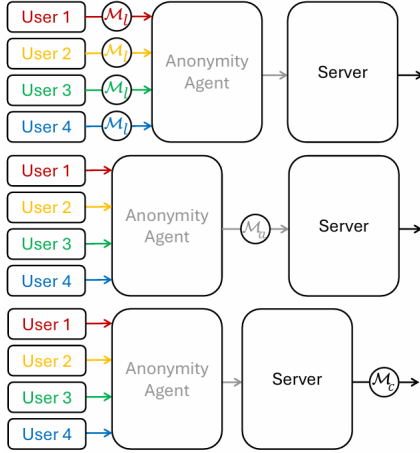


Figure 1: Comparison of privacy models, from top to bottom: Local Differential Privacy, Semi-Central Differential Privacy and Central Differential Privacy in multi-user systems

To incorporate privacy into MABs, we formalize protection over the individual reward feedback r_t , which may encode sensitive user information.

Differential Privacy (DP) is defined over two neighboring reward sequences $R = (r_1, \dots, r_T)$ and R' that differ in at most one element [14, 15]. A randomized algorithm $\mathcal{A}$ satisfies (ϵ, δ) -DP if for any subset S from the image of $\mathcal{A}$,

$$\Pr[\mathcal{A}(R) \in S] \leq e^\epsilon \Pr[\mathcal{A}(R') \in S] + \delta.$$

In the local differential privacy (LDP) model [44], each client locally perturbs their reward before sending it to the server using a DP mechanism $\mathcal{M}_l$.

In the Central DP model we assume that the learning process is trusted and because the server applies a DP mechanism $\mathcal{M}_c$ on the aggregation of the rewards, users do not need to privatize their rewards. The distinction is crucial: in Central DP, the server enforces privacy via noise addition to aggregate statistics, while in LDP, privacy is enforced before data collection, which is beneficial in adversarial or untrusted environments. We investigate these privacy models in the presence of an anonymity agent, which enforces the constraints required by the Anonymity model.

Motivated by the presence of the anonymity agent in these models, we introduce a novel differential privacy model called *Semi-Central DP*, in which we assume the anonymity agent is trusted, yet the environment and the server are not. So the anonymity agent would apply a DP mechanism $\mathcal{M}_a$ on the aggregation of users' rewards before sending their data to the server.

This new type of privacy model is similar to Local DP because our algorithm receives its inputs directly from the anonymity agent, so with respect to this agent, the algorithm can be seen locally differential private. However, the reward of the users fed to the anonymity agent is not changed (without applying any DP Mechanism), meaning that it is not completely the same as the Local DP model. Also it is similar to Central DP model as it applies DP mechanism to a function of rewards in order to reduce the regret term with respect to privacy. This new framework, which enforces privacy in a different way, bridges the local and Central DP model (in the presence of the anonymity agent) by providing a better regret with respect to the local and more privacy with respect to the Central DP model.

2.5 Assumptions

Here we describe all the necessary assumptions for our algorithms:

ASSUMPTION 1. We assume that $r_{t,i}(a_t) \in [0, 1]$ for all $t \in [T]$, $i \in [N]$, $a \in \mathcal{A}$.

This assumption is necessary and standard for providing concentration bounds.

ASSUMPTION 2. For every arm $a \in \mathcal{A}$, we assume there are at least $U \geq C + 1$ users for whom, a is the optimal arm.

This is a standard assumption imposed by [16]. They prove that this assumption is necessary in order to solve the problem and provide near-matching lower and upper bounds in the presence and not presence of this assumption.

3 Algorithms

In this section we intend to achieve a differentially private algorithm for Multi-User systems which preserves anonymity as well. Our algorithm has multiple components which work together to provide our privacy guarantees while suffering low regret. We will explain each of these components individually here.

3.1 Anonymous decompositions of bipartite graphs

Consider a weighted bipartite graph represented by a non-negative matrix $W = (w_{ij}) \in \mathbb{R}_+^{N \times K}$, where w_{ij} denotes the demand of user i for feedback from arm j . We assume $w_{ij} > 0$ for at least one arm j per user i and $D = \sum_{i,j} w_{ij}$ is the total demand.

A C -anonymous decomposition of W is a collection of R assignments (matchings) $M_1, \dots, M_R : [N] \rightarrow [K]$ such that:

- (1) Users are only assigned to arms for which they have positive demand, i.e., if $M_r(i) = j$ for any $r \in [R]$, then $w_{ij} > 0$.
- (2) For each user-arm pair (i, j) , there exist at least w_{ij} assignments M_r where i is matched to j and the number of users assigned to j in M_r is at least $C + 1$. Such assignments are called *informative* for (i, j) .

We focus on U -batched graphs, where each user i has a demand subset $A_i \subseteq [K]$, with demand $w_{ij} = \frac{D}{|A_i|}$ if $j \in A_i$ and zero otherwise, and each arm j has at least U users interested in it.

An anonymous decomposition algorithm takes a U -batched graph and outputs a C -anonymous decomposition with at most $\alpha D + O(NK)$ total assignments, where α is a function of K, C, U and N .

As in our algorithm, D will always be much larger than K (we will have $D \geq \sqrt{T}$), so $O(NK)$ can be thought of as an additive $o(D)$ term.

[16] proposes two different methods for constructing anonymous decompositions: a greedy method which achieves $\alpha = K$ and a combinatorial optimization-based method that guarantees $\alpha = \min(1, \lceil \frac{K(C+1)}{U} \rceil)$. We will assume we have access to a generic anonymous decomposition algorithm **Decompose** with approximation ratio α .

3.2 Feedback-eliciting subalgorithm

The Feedback-eliciting in [16] has a subtraction process for which the reward associated with the sum of each group's rewards is the same. This means that the estimators for each pair of users and arms are not necessarily independent, so it cannot provide the sensitivity required for privacy calculation. In order to solve these difficulties, we introduce a new version of this subroutine to be employed in our algorithm.

Formally, let

$$\pi : [N] \rightarrow [K]$$

be a matching from users to arms. We will show how to (in $4C + 2$ rounds) recover an unbiased estimate of $\mu_{i,\pi(i)}$ for each i such that

$$|\pi^{-1}(\pi(i))| \geq C + 1$$

(i.e. i is matched to an arm that at least C other users are matched to).

This algorithm receives and stores two rewards for each edge $(i, \pi(i))$ in assignment π . The value $g_{i,\pi(i)}$ represents the sum of at most $2C + 1$ samples corresponding to different users playing arm $\pi(i)$, including one sample from user i (i.e. $(i, \pi(i))$). In contrast, $h_{i,\pi(i)}$ denotes the sum of the remaining samples, which are independent fresh samples drawn from the same distributions as those contributing to $g_{i,\pi(i)}$, but excluding $(i, \pi(i))$.

It follows that $g_{i,\pi(i)} - h_{i,\pi(i)}$ serves as a random variable with the expectation of $\mu_{i,\pi(i)}$ (user i selecting arm $\pi(i)$) which includes the sum/subtraction of the corresponding noises of at most $4C + 1$ users. More concretely, we follow these steps:

- (1) For each arm j such that $|\pi^{-1}(j)| \geq C + 1$, partition the users in $\pi^{-1}(j)$ into groups of size at least $C + 1$ and at most $2C + 1$. Let $P_1, P_2, \dots, P_S$ be the set of groups formed in this way (over all arms j). In each group, order the users arbitrarily.
- (2) In the first $2C + 1$ rounds, the learner reports the partition into groups $\{P_1, P_2, \dots, P_S\}$. For each group P_s , devote the i -th round to the i -th user in P_s namely user $u_{s,i}$ and let $g_{u_{s,i},\pi(u_{s,i})}$ be the total aggregate reward for group P_s this round. If $i > |P_s|$ for some s , ignore the aggregated reward.
- (3) In the k th of the next $2C + 1$ rounds, the learner reports the partition into groups $\{P_{1,k}, P_{2,k}, \dots, P_{S,k}\}$, where $P_{s,k}$ is formed from P_s by removing the k -th element (if $k > |P_s|$, then we set $P_{s,k} = P_s$ and ignore the result). Let $h_{u_{s,i},\pi(u_{s,i})}$ be the total aggregate reward from $P_{s,k}$ reported this round.
- (4) If user i is the k th user in P_s , i.e. $i = u_{s,k}$ we have

$$\hat{\mu}_{i,\pi(i)} = g_{i,\pi(i)} - h_{i,\pi(i)}$$

and we return $g_{i,\pi(i)}, h_{i,\pi(i)}$.

Algorithm 1 Batched Successive Elimination (BaSE)

Input: Number of arms K , number of batches B , accuracy parameters γ_τ .

- 1: Set $A = [K]$ representing remaining arms.
 - 2: $r_i, n_i = 0$ for $i \in A$.
 - 3: **for** $b = 1, 2, \dots, B - 1$ **do**
 - 4: Receive y_i, m_i for $i \in A$ from the superior algorithm (representing the total reward from m_i pulls of arm i).
 - 5: $r_i \leftarrow r_i + y_i, n_i \leftarrow n_i + m_i$ for $i \in A$.
 - 6: $\hat{\mu}_{i,b} \leftarrow r_i / n_i$ for $i \in A$.
 - 7: Let $a^* = \arg \max_{a \in A} r_a / n_a$.
 - 8: Let $\tau_b = \min_{i \in A} n_i$.
 - 9: Remove any arm i from A if $\hat{\mu}_{a^*,b} - \hat{\mu}_{i,b} > 2\gamma_{\tau_b}$.
 - 10: **end for**
-

In the above procedure, $\mathbb{E}[\hat{\mu}_{i,\pi(i)}] = \mu_{i,\pi(i)}$ and $\hat{\mu}_{i,\pi(i)} - \mu_{i,\pi(i)}$ is the sum of at most $4C + 1$ corresponding noises, which includes independent sub-Gaussian noises in addition to Laplace noises based on the DP model.

3.3 BaSE

First we introduce a DP-adaptable modified BaSE (Batch Successive Elimination) algorithm [18]. In this algorithm, we use new samples of rewards of active arms and an updated accuracy parameter (which is designed for each batch efficiently) from a superior algorithm and output the remaining arms which are still candidates for being the best arm.

This algorithm initializes the arms in first steps and after that it will receive $B - 1$ set of rewards and for each step, it will calculate the empirical mean of active arms and remove sub-optimal arms for the next batch.

3.4 ADPBaSE

ADPBaSE uses N instances of BaSE in order to propose a differentially private algorithm for Anonymous Multi-User systems. The inputs of this algorithm are received from the anonymity agent by Feedback-eliciting subalgorithm. The anonymity agent plays a single action for a group of users and applies anonymous privacy by aggregating the rewards of users and returns the result only if the number of users in the group are more than the desired anonymity parameter C .

Based on the desired DP-Model, one of the components adds the required Laplace noise on its output. In order to provide ϵ -differential privacy, either users directly add noise on their rewards, or the anonymity agent does the same on the reward aggregations or learner on the aggregation of all rewards for each batch.

We create a new time horizon T' , in each new time step we use $\alpha(4C + 2)$ steps for decomposition and Feedback-eliciting sub algorithms. In each batch, we feed the new rewards to these BaSE instances and we receive new demand vectors from each instance for every user. By running decomposition on these demand vectors and using Feedback-eliciting on the resulting assignments, we will be guaranteed to have the required number of samples for each active arm for every user.

Algorithm 2 Anonymized Differentially Private Batched Successive Elimination (ADPBaSE)

Input: Number of users N , number of arms K , time horizon T , number of batches B , privacy parameters ϵ , accuracy parameter γ_τ , anonymity parameter C .

- 1: For each user $i \in [N]$, initialize BaSE _{i} with adjusted horizon $T' = T/\alpha(4C + 2)$, $\log \log T'$ batches, and accuracy parameter γ_τ .
- 2: Set batch sizes $D_1, D_2, \dots, D_B$ over T' as in [18].
- 3: Initialize active arms $A_{1,1}, \dots, A_{N,1} = [K]$.
- 4: **for** $b = 1$ **to** B **do**
- 5: Run **Decompose** on the U -graph created by $A_{1,b}, \dots, A_{N,b}$ and total demand D_b to obtain C -anonymous assignments $M_1, \dots, M_{R_b}$ where $R_b = \alpha D_b$.
- 6: **for all** $i \in [N], j \in [K]$ **do**
- 7: Initialize $\bar{g}_{i,j,b} \leftarrow 0, \bar{h}_{i,j,b} \leftarrow 0, m_{i,j} \leftarrow 0$.
- 8: **end for**
- 9: **for** $r = 1$ **to** R_b **do**
- 10: Run Feedback-eliciting subalgorithm to get $g_{i,j}, h_{i,j}$ for each (i, j) .
- 11: Update: $\bar{g}_{i,j,b} += g_{i,j}, \bar{h}_{i,j,b} += h_{i,j}, m_{i,j} += 1$.
- 12: **end for**
- 13: **for** $i = 1$ **to** N **do**
- 14: **for** $j = 1$ **to** K **do**
- 15: If in Central DP Model, add noise: $\bar{g}_{i,j,b} \leftarrow \bar{g}_{i,j,b} + z_{i,j,b}, \bar{h}_{i,j,b} \leftarrow \bar{h}_{i,j,b} + z'_{i,j,b}$, where $z_{i,j,b}, z'_{i,j,b} \sim \text{Lap}(1/\epsilon)$.
- 16: Compute: $y_{i,j,b} = \bar{g}_{i,j,b} - \bar{h}_{i,j,b}$ and set $m_{i,j,b} = m_{i,j}$.
- 17: **end for**
- 18: Feed the set $\{(y_{i,j,b}, m_{i,j,b}) : j \in [K]\}$ to BaSE _{i} , and create $A_{i,b+1}$ using BaSE _{i} 's set of active arms if $b < B$.
- 19: **end for**
- 20: **end for**

4 Analysis

In this section we provide upper bounds of the algorithm for each model. First we use these definitions and lemmas to ease our explanations and proofs:

DEFINITION 1. Let $\hat{\mu}_{i,n,b}$ and $\tau_{b,n}$ denote $\hat{\mu}_{i,b}$ and τ_b in the n -th BaSE instance in algorithm 1, respectively.

DEFINITION 2. We assume that $t_b = \sum_{i=1}^b D_i$ denotes the number of rounds up to the last round of batch i , where D_i is the i -th batch size in algorithm 2.

Based on this uniform selection of arms, we can provide a lower bound for $\tau_{b,n}$ in each instance of algorithm 1 in all models, which will be useful in regret analysis.

LEMMA 1. During algorithm 2 in all three models of privacy, based on uniform pulls of arms we always have that $\tau_{b,n} \geq t_b/K - b$.

PROOF. We get at least $\lfloor \frac{D_i}{K} \rfloor$ independent samples in batch i , which is greater than $\frac{D_i}{K} - 1$. Now summing up on all batches will prove the lemma. $\square$

If in any step we remove the best arm, we cannot bound the regret due to the fact that we are comparing some sub-optimal arms, so we need to make sure the optimal arm is always there.

LEMMA 2. During algorithm 2, if for all $i \in \mathcal{A}, b \in [B], n \in [N]$

$$|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b},$$

then for each user their optimal arm will not be removed.

PROOF. Suppose that the optimal arm for some user n has been removed during batch b in step 9 of algorithm 1, so there is an arm j with $\hat{\mu}_{j,b,n} - \mu_{j,n}^* > 2\gamma_{\tau_{b,n}}$, so

$$\mu_{j,n} \geq \hat{\mu}_{j,b,n} - \gamma_{\tau_{b,n}} > \mu_{b,n}^* + \gamma_{\tau_{b,n}} \geq \mu_n^*,$$

which is a contradiction due to the fact that $\mu_n^* \geq \mu_{j,n}$. $\square$

Now with optimal arm always available in the action set we can bound each steps regret.

LEMMA 3. During algorithm 2, if for all $i \in \mathcal{A}, b \in [B], n \in [N]$

$$|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b},$$

then in each step during batch $b > 0$ we suffer a regret of at most

$$4\gamma_{\tau_{b-1}}.$$

PROOF. Suppose that $\hat{\mu}_j$ be the estimation of the best estimated arm in the previous batch for any user. Based on lemma 2 the optimal arm has not been eliminated, so we have that

$$\mu^* - \gamma_{\tau_{b-1}} \leq \hat{\mu}^* \leq \hat{\mu}_j.$$

So for any remaining arm i we have that

$$\hat{\mu}_j < \hat{\mu}_i + 2\gamma_{\tau_{b-1}} \leq \mu_i + 3\gamma_{\tau_{b-1}}$$

based on the algorithm's removing policy and we can say that

$$\mu^* - \gamma_{\tau_{b-1}} \leq \mu_i + 3\gamma_{\tau_{b-1}} \Rightarrow \mu^* - \mu_i \leq 4\gamma_{\tau_{b-1}}.$$

So in each action during this batch we have the regret of at most $4\gamma_{\tau_{b-1}}$. $\square$

4.1 Local DP Model

In the Local DP Model, we suppose that our anonymity server is not trusted and each user needs to provide privacy for their input using a DP mechanism $\mathcal{M}_l$.

We use Laplace noise as this mechanism, in which all users apply a Laplace noise

$$z \sim \text{Lap}\left(\frac{1}{\epsilon}\right)$$

on their own reward and send it to the anonymity agent.

Regret Analysis. To show that our accuracy parameter performs well in this privacy setting, we need to bound the difference between the empirical estimation and the true value of the actions. In the following lemma, we provide such a bound:

LEMMA 4. During ADPBaSE, by setting the accuracy parameter

$$\gamma_\tau = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau}} + \frac{1}{\epsilon} \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau}}.$$

and $\delta = 1/T$, we have for all $i \in \mathcal{A}, b \in [B], n \in [N]$ that

$$|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$$

with probability less than 2δ .

Using this model of applying Laplace noises, we can prove that algorithm 2 is ϵ -LDP and also we can provide a high probability bound on its regret.

THEOREM 1. *By choosing the accuracy parameter according to lemma 4 and $\delta = 1/T$, ADPBaSE is ϵ -LDP and will suffer a regret of at most*

$$\tilde{O}\left(\frac{NC}{\epsilon}\sqrt{\alpha KT}\right).$$

with probability at least $1 - 2/T$.

PROOF. Assume that $|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$ holds for all i, n, b . First we calculate the regret in batch $b > 1$ for one instance of BaSE. Based on 3 in each action during this batch we have the regret of at most $4\gamma_{\tau_{b-1}}$. Now we try to bound $\gamma_{\tau_{b-1}}$:

$$\begin{aligned} \gamma_{\tau_{b-1}} &= \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} + \frac{1}{\epsilon} \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} \\ &= (1 + \frac{1}{\epsilon}) \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} \\ &\leq (1 + \frac{1}{\epsilon}) \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K - b}} \quad (1) \\ &\approx (1 + \frac{1}{\epsilon}) \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} \quad (2) \end{aligned}$$

where approximation 1 holds based on lemma 1 and $b \leq B = \log \log T'$ and $t_{b-1} = \Omega(\sqrt{T'})$ so $t_{b-1} \gg b$ and the approximation 2 holds. Now using this bound and the fact that in this batch we select $\alpha(4C + 2)D_b$ arms each with sub-optimality of at most $4\gamma_{\tau_{b-1}}$, we bound the regret for b -th batch

$$\begin{aligned} R_b &= 4\alpha(4C + 2)D_b \gamma_{\tau_{b-1}} \\ &\leq 4\alpha p C t_b (1 + \frac{1}{\epsilon}) \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} \quad (3) \end{aligned}$$

$$\begin{aligned} &= 4\alpha p C a \sqrt{t_{b-1}} (1 + \frac{1}{\epsilon}) \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} \quad (4) \\ &= 4\alpha p C a (1 + \frac{1}{\epsilon}) \sqrt{2pCK \log(\frac{2KNB}{\delta})} \end{aligned}$$

where inequality 3 holds because $t_b \geq D_b$ and $pC \geq (4C + 2)$. Equality 4 holds based on the assignments of batch sizes in [18] and $a = \Theta(T'^{\frac{1}{2-2^{1-M}}})$. As

$$T' = T/\alpha(4C + 2) \leq T/\alpha C$$

we can say that

$$\begin{aligned} R_b &= O(4\alpha p C \sqrt{T/\alpha C} (1 + \frac{1}{\epsilon}) \sqrt{2pCK \log(\frac{2KNB}{\delta})}) \\ &= O(\frac{C}{\epsilon} \sqrt{\alpha KT \log(\frac{2KNB}{\delta})}) \end{aligned}$$

for $b > 1$.

In first batch we suffer regret of at most 1 for each action, so

$$\begin{aligned} R_1 &= O(4\alpha p C a) \\ &= O(4\alpha p C \sqrt{T/\alpha C}) \\ &= O(\sqrt{T\alpha C}). \end{aligned}$$

Now we can say that

$$\mathcal{R}_T = \sum_{b=1}^B R_b = O\left(\frac{BC}{\epsilon} \sqrt{\alpha KT \log(\frac{2KNB}{\delta})}\right)$$

and the total regret of all instances will be

$$O\left(\frac{NBC}{\epsilon} \sqrt{\alpha KT \log(\frac{2KNB}{\delta})}\right).$$

Now based on lemma 4 the condition $|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$ holds with probability $1 - 2/T$, so the total regret would be

$$O\left((1 - \frac{2}{T}) \frac{NBC}{\epsilon} \sqrt{\alpha KT \log(\frac{2KNB}{\delta})} + \frac{2}{T} TN\right).$$

As $B = \log \log T'$ the total regret would be

$$\tilde{O}\left(\frac{NBC}{\epsilon} \sqrt{\alpha KT}\right)$$

and we have proven the theorem. $\square$

Assuming $U \geq K(C+1)$ we can achieve a regret of $\tilde{O}(NC\sqrt{KT}/\epsilon)$ which matches the lower bound of [16] up to a factor of $\sqrt{C}$ without privacy ($\epsilon \geq 1$) and a loss from privacy $1/\epsilon$ which for small ϵ matches the lower bound of $\frac{\sqrt{T}}{\exp(\epsilon)-1}$ in [6] in T, ϵ terms since for small enough ϵ one could say $\epsilon \approx \exp(\epsilon) - 1$.

Privacy Analysis. Since for any user the reward is bounded by $0 \leq r \leq 1$, the output $r + z$, $z \sim \text{Lap}(\frac{1}{\epsilon})$ is ϵ -LDP from [14].

4.2 Central DP Model

We assume that the whole learning process and server (centralized learner) are trusted. Thus, the trusted server can apply a DP mechanism $\mathcal{M}_c$ to preserve the privacy of the collected rewards. The learner uses a Laplace noise in algorithm 2 line 15 to ensure privacy.

Regret Analysis. First we try to bound the difference of the empirical and real means.

LEMMA 5. *During ADPBaSE, by setting the accuracy parameter*

$$\gamma_{\tau} = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau}} + \frac{\log(\frac{2KNB}{\delta}) + B}{\epsilon \tau}$$

and $\delta = 1/T$, we have for all i, b, n that

$$|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$$

with probability less than 2δ .

With both anonymity agent and learner being trusted, our algorithm provides ϵ -differential privacy while suffering a high probability low regret.

THEOREM 2. By choosing the accuracy parameter according to lemma 5 and $\delta = 1/T$, ADPBaSE is ϵ -Differentially Private and will suffer a regret of at most

$$\tilde{O}(NC\sqrt{TK\alpha} + \frac{N(C\alpha)^{\frac{3}{4}}T^{\frac{1}{4}}K}{\epsilon})$$

with probability at least $1 - 2/T$.

If $U \geq K(C + 1)$, then $\alpha = 1$ and our algorithm achieves a regret bound of

$$\tilde{O}(NC\sqrt{TK} + NC^{3/4}T^{1/4}K/\epsilon).$$

We can see that if $T > K^2/C\epsilon^4$ (which is true as T grows) then the second term in regret is smaller than $\tilde{O}(NC\sqrt{TK})$ and our algorithm will achieve differential privacy without cost in regret, matching the best known algorithm [16].

Privacy Analysis. Let $\bar{g} = [\bar{g}_{i,j,b}]$ and $\bar{h} = [\bar{h}_{i,j,b}]$ for $i \in [N]$, $j \in [K]$, $b \in [B]$ and let $\hat{g}$ and $\hat{h}$ be $\bar{g}, \bar{h}$ after applying Laplace noises. Also let $\bar{r} = [\bar{g}, \bar{h}]$ and $\hat{r} = [\hat{g}, \hat{h}]$.

Now consider two neighboring sequences of rewards R, R' . If $\bar{r}$ and $\bar{r}'$ be corresponding to these reward sequences, it is obvious that

$$\|\bar{r} - \bar{r}'\|_1 \leq 1.$$

due to the fact that each r_t appears in both $\bar{r}$ and $\bar{r}'$ once except their one difference. Now we can say that based on [14], $\hat{r}$ is ϵ -DP. As our algorithm has access to R only through $\hat{r}$ and the output of algorithm 2 deterministically depends on R , by post processing algorithm 2 is ϵ -DP.

4.3 Semi-Central DP Model

In this model, the anonymity agent who collects the rewards from users is trusted but the centralized learner (server) and the subsequent learning process is not. Users' rewards are sent to the anonymity agent and the agent applies a DP mechanism $\mathcal{M}_a$ on them, which is a Laplace noise $z \sim \text{Lap}(\frac{1}{\epsilon})$.

This model achieves better privacy guarantees than the Central DP model and better regret bounds than the Local DP model.

Regret analysis. To bound the difference of empirical mean and the value of arms, we use the following lemma:

LEMMA 6. During ADPBaSE By setting the accuracy parameter

$$\gamma_\tau = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau}} + \frac{1}{\epsilon} \sqrt{\frac{4 \log(\frac{2KNB}{\delta})}{\tau}}$$

and $\delta = 1/T$, we have for all i, b, n that

$$|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$$

with a probability less than 2δ .

Now we can prove that with a trusted anonymity agent, our algorithm is ϵ -DP and has low regret with high probability.

THEOREM 3. By choosing the accuracy parameter according to lemma 6 and $\delta = 1/T$, ADPBaSE is ϵ -Differentially Private and will suffer a regret of at most

$$\tilde{O}(N(\sqrt{C} + \frac{1}{\epsilon})\sqrt{\alpha CKT})$$

with probability at least $1 - 2/T$.

By setting $\epsilon = \frac{1}{\sqrt{C}}$ the proposed method has a regret equal to the best known regret of the anonymous model without differential privacy [16]. Also if $\frac{1}{\epsilon} \gg \sqrt{C}$, this model has improved the regret of the Local DP model by a term of $\sqrt{C}$.

Privacy analysis. We receive $T'\alpha$ feedbacks from the anonymity agent. Let $\bar{r} = [r_{P_1}, \dots, r_{P_{T'\alpha}}]$ be the rewards collected by this agent for each step and suppose that $\hat{r} = [r_{P_1} + z_1, \dots, r_{P_{T'\alpha}} + z_{T'\alpha}]$ are noisy rewards where

$$z_i \sim \text{Lap}(\frac{1}{\epsilon}).$$

Consider two neighboring sequences of rewards R and R' , which only differ in r_t and r'_t . Also suppose that $\bar{r}, \bar{r}'$ be the rewards collected by the anonymity agent based on R, R' . As we only use the summation function for the Anonymity Agent, each reward in $\bar{r}$ appears in $\bar{r}'$ except r_t , so:

$$\|\bar{r} - \bar{r}'\|_1 \leq \max_{r_t, r'_t} |r_t - r'_t| \leq 1$$

where the last inequality follow from the fact that rewards are bounded by 1. Now we can say that based on [14], $\hat{r}$ is ϵ -DP.

As our algorithm has access to R only through $\hat{r}$ and the output of algorithm 2 deterministically depends on R , by post processing algorithm 2 in Semi-Central DP model is ϵ -DP.

5 Conclusion

This work presents the first unified framework combining differential privacy with anonymity in multi-user bandit systems. We develop ADPBaSE algorithms achieving regret bounds of

$$\tilde{O}(NC\sqrt{KT\alpha}/\epsilon)$$

under the Local DP model,

$$\tilde{O}(NC\sqrt{KT\alpha} + N(C\alpha)^{3/4}T^{1/4}K/\epsilon)$$

under Central DP model, and introduce an intermediate Semi-Central DP model with

$$\tilde{O}(NC\sqrt{KT\alpha} + N\sqrt{CKT\alpha}/\epsilon)$$

regret that bridges Local DP and Central DP approaches.

Our Semi-Central DP model offers a practical compromise by trusting anonymity aggregation while maintaining privacy against the learning server.

These results demonstrate that strong privacy guarantees and effective collaborative learning can coexist, enabling deployment in sensitive applications like healthcare and personalized recommendations where both anonymity and differential privacy are required.

Since aggregation naturally obscures individual contributions when users have similar reward distributions, it adapts to the reward structure. Thus anonymity complements DP, improving interpretability and adding protection via the structure of aggregated samples. The framework provides theoretical foundations for next-generation privacy-preserving machine learning systems.

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This appendix is organized as follows:

Section A presents additional related work that complements the discussion in the main paper. Section B provides detailed proofs for the Local DP model. Section C contains the formal analysis and proofs for the Central DP model. Section D offers theoretical results and proofs for the Semi-Central DP model.

A Other Related Works

Multi-armed bandit problems have emerged as a fundamental framework in sequential decision-making, attracting significant research attention across disciplines including machine learning, statistics, and operations research [2, 3, 9, 29].

Subsequent research has applied the batching paradigm to other domains, such as Thompson Sampling [21, 22], Gaussian process bandits [25], and contextual bandits [42, 43]. Moreover, the convex bandit variant has also been studied in batch-constrained settings [13], though primarily from a best-arm identification perspective. The theoretical analysis of batch-constrained learning connects to broader questions in computational complexity theory. Parallel algorithm design has long grappled with similar tradeoffs between computational rounds and solution quality [1, 7, 38]. More

recent work has extended these ideas to noisy or stochastic settings [8, 11, 17], developing techniques that prove relevant to bandit analysis.

Meanwhile, privacy-preserving learning in bandit settings has been actively studied through the lens of differential privacy. Algorithms that add noise to either the action selection process or observed rewards have been proposed to protect user-level data [6, 12, 33, 37]. These approaches typically focus on single-user environments and assume access to individual-level feedback, which limits their applicability in scenarios demanding strict anonymity.

B Local DP Model

First we intend to use concentration bounds on noises to bound them with high probability. In the following lemma, we use a famous Chernoff bound for sub-Gaussian noises to bound sum of independent sub-Gaussian noises.

LEMMA 7. Let $\bar{w} = \frac{\sum_{i=1}^n w_i}{\tau}$, where w_i follows a 1-sub-Gaussian distribution. We have

$$\Pr[|\bar{w}| \geq q] \leq 2 \exp\left(-\frac{q^2 \tau^2}{2n}\right)$$

Now we need to bound the average of independent Laplace noises. In order to do that we use the following lemma from [19]:

LEMMA 8. Let $x_i = l_i z_i$ for $i \in [n]$, where $z_i \sim \text{Lap}(\frac{1}{b})$ and l_i is constant such that $|l_i| \leq c$. Let $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$. We have that

$$\Pr[\bar{x} \geq t] \leq \begin{cases} \exp(-\frac{t^2 b^2}{2nc^2}) & \text{if } t \leq \frac{nc}{b} \\ \exp(\frac{n}{2} - \frac{b}{c}t) & \text{if } t > \frac{nc}{b}. \end{cases}$$

We use a simpler version of lemma 8 for our analysis.

LEMMA 9. Let $\bar{z} = \frac{\sum_{i=1}^n z_i}{\tau}$, where $z_i \sim \text{Lap}(\frac{1}{b})$. We have that

$$\Pr[|\bar{z}| \geq q] \leq \begin{cases} 2 \exp(-\frac{q^2 \epsilon^2 \tau^2}{2n}) & \text{if } q \leq \frac{n}{\epsilon \tau} \\ 2 \exp(\frac{n}{2} - q \epsilon \tau) & \text{if } q > \frac{n}{\epsilon \tau} \end{cases}$$

PROOF. Direct result of Lemma 8 by setting $C = \frac{1}{\tau}$ and $b = \epsilon$ and $t = q$ and a union bound on $\Pr[\bar{z} \geq q]$ and $\Pr[\bar{z} \leq -q]$ and the fact that our noises are symmetrical and zero mean so $\Pr[\bar{z} \geq q] = \Pr[\bar{z} \leq -q]$. $\square$

Now we can bound the difference of our empirical means and the real means of arms using lemmas presented above.

Proof of lemma 4.

PROOF. We know that $\hat{\mu}_{i,n,b} = \mu_{i,n} + \bar{w} + \bar{z}$ where

$$\bar{w} = \frac{\sum_{i=1}^s w_i}{\tau_0}$$

where the w_i 's are 1 sub-Gaussian noises and

$$\bar{z} = \frac{\sum_{i=1}^s z_i}{\tau_0}$$

where $z_i \sim \text{Lap}(\frac{1}{\epsilon})$ and τ_0 indicates the number of samples we received for this arm and user n . Also it is obvious that $\tau_0 \geq \tau_{b,n}$ based on the selection of $\tau_{b,n}$. We know that for each sample we have at most $4C + 1$ independent noises based on Feedback-eliciting subalgorithm, so $s \leq pC\tau_0$ where p is a constant number. Now we

want to calculate the probability of $|\mu_{i,n} - \hat{\mu}_{i,n,b}| \leq \gamma_{\tau_{b,n}}$. Suppose that $u = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b,n}}}$ and $v = \frac{1}{\epsilon} \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{n,b}}}$. Now based on $\gamma_{\tau_{b,n}} = u + v$ and $|\mu_{i,n} - \hat{\mu}_{i,n,b}| = |\bar{w} + \bar{z}|$ we want to find

$$\Pr[|\bar{w} + \bar{z}| \geq u + v].$$

Also,

$$\Pr[(|\bar{w}| \geq u) \cup (|\bar{z}| \geq v)] \geq \Pr[|\bar{w} + \bar{z}| \geq u + v]$$

because at least one of $|\bar{w}| \geq u$ and $|\bar{z}| \geq v$ needs to be true in order to $|\bar{w} + \bar{z}| \geq u + v$. Now based on union bound inequality we have that

$$\Pr[(|\bar{w}| \geq u)] + \Pr[(|\bar{z}| \geq v)] \geq \Pr[(|\bar{w}| \geq u) \cup (|\bar{z}| \geq v)].$$

We want to prove that each of $\Pr[(|\bar{w}| \geq u)]$ and $\Pr[(|\bar{z}| \geq v)]$ are less than $\frac{2\delta}{KNB}$. Now

$$\Pr[(|\bar{w}| \geq u)] \leq 2 \exp\left(-\frac{u^2 \tau_0^2}{2s}\right) \quad (5)$$

$$\begin{aligned} &= 2 \exp\left(-\left(\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{n,b}}\right) \frac{\tau_0^2}{2s}\right) \\ &\leq 2 \exp\left(-\left(\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_0}\right) \frac{\tau_0^2}{2pC\tau_0}\right) \\ &= 2 \exp\left(-\log\left(\frac{2KNB}{\delta}\right)\right) \\ &= \frac{\delta}{KNB}, \end{aligned} \quad (6)$$

where inequality 5 is based on lemma 7 and inequality 6 is based on the fact that $\tau_{b,n} \leq \tau_0$ and $s \leq pC\tau_0$. Also, we have that

$$\Pr[(|\bar{z}| \geq v)] \leq 2 \exp\left(-\frac{v^2 \epsilon^2 \tau_0^2}{2s}\right) \quad (7)$$

$$\begin{aligned} &= 2 \exp\left(-\left(\frac{2pC \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_{n,b}}\right) \frac{\epsilon^2 \tau_0^2}{2s}\right) \\ &\leq 2 \exp\left(-\left(\frac{2pC \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_0}\right) \frac{\epsilon^2 \tau_0^2}{2pC\tau_0}\right) \\ &= 2 \exp\left(-\log\left(\frac{2KNB}{\delta}\right)\right) \\ &= \frac{\delta}{KNB}, \end{aligned}$$

where the inequality 7 is a direct result of lemma 9 if $v \leq \frac{s}{\epsilon \tau_0}$, which is true because based on the grid and size of the first batch we have that $\tau_{n,b}$ is at least $O(\sqrt{T'}/K)$, and $T' \gg C, \log(KNB/\delta), K$, so $v \leq \frac{1}{\epsilon} \leq \frac{s}{\tau_0 \epsilon}$ ($s \geq \tau_0$). Now, we proved that

$$\begin{aligned} \Pr[|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_{b,n}}] &= \Pr[|\bar{w} + \bar{z}| \geq u + v] \\ &\leq \Pr[(|\bar{w}| \geq u)] + \Pr[(|\bar{z}| \geq v)] \\ &\leq \frac{2\delta}{KNB}. \end{aligned}$$

Now, by using a union bound on all arms $i \in [K]$, all users $n \in [N]$ and all batches $b \in [B]$ the theorem is proven. $\square$

C Central DP Model

We will follow the same approach of proving the regret bound as the Local DP model.

Proof of lemma 5.

PROOF. We have $\hat{\mu}_{i,n,b} = \mu_{i,n} + \bar{w} + \bar{z}$ where

$$\bar{w} = \frac{\sum_{i=1}^s w_i}{\tau_0}$$

where the w_i 's are 1 sub-Gaussian noises and

$$\bar{z} = \frac{\sum_{i=1}^{2b} z_i}{\tau_0}$$

where $z_i \sim \text{Lap}(\frac{1}{\epsilon})$. τ_0 indicates the number of samples we received for this arm and user n and $\tau_0 \geq \tau_{b,n}$. We know that for each sample we have at most $4C + 1$ independent 1-sub-Gaussian noises based on Feedback-eliciting subalgorithm, so $s \leq pC\tau_0$ where p is a constant number. Also the algorithm applies Laplace noises on each batch's aggregation of rewards y_b and g_b , so we have $2b$ independent Laplace noises.

Like the proof of Theorem 1 we set $u = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b,n}}}$ and $v = \sqrt{\frac{4 \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_{b,n}}}$ and we know that

$$\Pr[|\bar{w}| \geq u] + \Pr[|\bar{z}| \geq v] \geq \Pr[|\bar{w} + \bar{z}| \geq u + v]$$

using union bound inequality.

As u, s are the same as u, s in the proof of theorem 1, we can follow the same steps and say

$$\Pr[|\bar{w}| \geq u] \leq \delta / KNB.$$

For $\bar{z}$ we have that

$$\Pr[|\bar{z}| \geq v] \leq 2 \exp\left(\frac{2b}{2} - v\epsilon\tau_0\right) \quad (8)$$

$$\begin{aligned} &= 2 \exp\left(b - \left(\frac{\log(\frac{2KNB}{\delta}) + B}{\epsilon\tau_{b,n}}\right)\epsilon\tau_0\right) \\ &\leq 2 \exp\left(b - \left(\frac{\log(\frac{2KNB}{\delta}) + b}{\epsilon\tau_0}\right)\epsilon\tau_0\right) \quad (9) \\ &= 2 \exp\left(-\log\left(\frac{2KNB}{\delta}\right)\right) \\ &= \frac{\delta}{KNB} \end{aligned}$$

where the inequality 8 is a result of lemma 9 if $v > \frac{b}{\epsilon\tau_0}$, which is true because

$$\begin{aligned} v &= \frac{\log(\frac{2KNB}{\delta}) + B/2}{\epsilon\tau_{b,n}} > \frac{\log(\frac{2KNB}{\delta})}{\epsilon\tau_{b,n}} \\ &\geq \frac{\log(T)}{\epsilon\tau_{b,n}} \geq \frac{\log(T)}{\epsilon\tau_0} > \frac{b}{\epsilon\tau_0}. \end{aligned}$$

Where the second inequality holds because $\frac{\log(\frac{2KNB}{\delta})}{\epsilon\tau_{b,n}} > \frac{\log(\frac{1}{\delta})}{\epsilon\tau_{b,n}} = \frac{\log(T)}{\epsilon\tau_{b,n}} > \frac{\log(T)}{\epsilon\tau_0}$. The third one holds because $\tau_{b,n} \leq \tau_0$ the fourth one holds because $b \leq B = \log \log T$. So we can write

$$\begin{aligned} \Pr[|\mu_{i,n} - \hat{\mu}_{i,n,b}| \leq \gamma_{\tau_{b,n}}] &= \Pr[|\bar{w} + \bar{z}| \geq u + v] \\ &\leq \Pr[|\bar{w}| \geq u] + \Pr[|\bar{z}| \geq v] \\ &\leq \frac{2\delta}{KNB} \end{aligned}$$

Now by using a union bound on all arms $i \in [K]$, all users $n \in [N]$ and all batches $b \in [B]$ the theorem is proved. $\square$

Proof of Theorem 2.

PROOF. We follow the same approach we used during the proof of Theorem 1. Assuming $|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$ for all i, n, b , based on 3 in each action during this batch we have the regret of at most $4\gamma_{\tau_{b-1}}$. Now we try to bound $\gamma_{\tau_{b-1}}$:

$$\begin{aligned} \gamma_{\tau_{b-1}} &= \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} + \frac{\log(\frac{2KNB}{\delta}) + B}{\epsilon\tau_{b-1}} \\ &\leq \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K - b}} + \frac{\log(\frac{2KNB}{\delta}) + B}{\epsilon(t_{b-1}/K - b)} \quad (10) \end{aligned}$$

$$\begin{aligned} &\approx \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} + \frac{\log(\frac{2KNB}{\delta}) + B}{\epsilon t_{b-1}/K} \\ &\approx \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} + \frac{\log(\frac{2KNB}{\delta})}{\epsilon t_{b-1}/K} \quad (11) \end{aligned}$$

where inequality 10 holds based on lemma 1 and $B = \log \log T$ and $\log(\frac{2KNB}{\delta}) \gg \log \log T$, 11 holds. Now using this bound and the fact that in this batch we select $\alpha(4C + 2)D_b$ arms each with the sub-optimality of at most $4\gamma_{\tau_{b-1}}$, we bound the regret for b -th batch

$$\begin{aligned} R_b &= 4\alpha(4C + 2)D_b\gamma_{\tau_{b-1}} \\ &\leq 4\alpha pCt_b \left(\sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} + \frac{\log(\frac{2KNB}{\delta})}{\epsilon t_{b-1}/K} \right) \quad (12) \\ &= 4\alpha pCa\sqrt{t_{b-1}} \left(\sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} + \frac{\log(\frac{2KNB}{\delta})}{\epsilon t_{b-1}/K} \right) \\ &= 4\alpha pCa \left(\sqrt{2pCK \log(\frac{2KNB}{\delta})} + \frac{K \log(\frac{2KNB}{\delta})}{\epsilon \sqrt{t_{b-1}}} \right) \end{aligned}$$

where inequality 12 holds because $t_b \geq D_b$ and $pC \geq (4C + 2)$. As $T' = T/\alpha(4C + 2) \leq T/\alpha C$ and $a = \Theta(T'^{\frac{1}{2-2^{1-M}}})$ we have

$$\begin{aligned} R_b &= O(\alpha Ca \sqrt{CK \log(\frac{2KNB}{\delta})} + \frac{\alpha CaK \log(\frac{2KNB}{\delta})}{\epsilon \sqrt{t_{b-1}}}) \\ &= O(\alpha C \sqrt{T/\alpha C} \sqrt{CK \log(\frac{2KNB}{\delta})} + \frac{\alpha CaK \log(\frac{2KNB}{\delta})}{\epsilon \sqrt{t_{b-1}}}) \\ &\leq O(C \sqrt{TK \alpha \log(\frac{2KNB}{\delta})} + \frac{\alpha CaK \log(\frac{2KNB}{\delta})}{\epsilon \sqrt{t_1}}) \\ &= O(C \sqrt{TK \alpha \log(\frac{2KNB}{\delta})} + \frac{\alpha CaK \log(\frac{2KNB}{\delta})}{\epsilon \sqrt{a}}) \\ &= O(C \sqrt{TK \alpha \log(\frac{2KNB}{\delta})} + \frac{\alpha C(T/\alpha C)^{\frac{1}{4}} K \log(\frac{2KNB}{\delta})}{\epsilon}) \end{aligned}$$

for $b > 1$. In first batch we suffer regret of at most 1 for each action, so

$$\begin{aligned} R_1 &= O(4\alpha pCa) \\ &= O(4\alpha pC \sqrt{T/\alpha C}) \\ &= O(\sqrt{TaC}). \\ &\leq R_b \end{aligned} \quad (13)$$

Now we can say that

$$\sum_{b=1}^B R_b = O(BC\sqrt{TK\alpha \log(\frac{2KNB}{\delta})} + \frac{B(C\alpha)^{\frac{3}{4}}T^{\frac{1}{4}}K \log(\frac{2KNB}{\delta})}{\epsilon})$$

and the total regret of all instances will be

$$\tilde{O}((1 - \frac{2}{T})(NBC\sqrt{TK\alpha} + \frac{NB(C\alpha)^{\frac{3}{4}}T^{\frac{1}{4}}K}{\epsilon}) + \frac{2}{T}NT)$$

As $B = \log \log T'$ we have proved the theorem. $\square$

D Semi-Central DP Model

Proof of lemma 6.

PROOF. We know that $\hat{\mu}_{i,n,b} = \mu_{i,n} + \bar{w} + \bar{z}$ where

$$\bar{w} = \frac{\sum_{i=1}^s w_i}{\tau_0}$$

where the w_i s are 1-sub-Gaussian noises. Also

$$\bar{z} = \frac{\sum_{i=1}^l z_i}{\tau_0}$$

where $z_i \sim \text{Lap}(\frac{1}{\epsilon})$. τ_0 indicates the number of samples received for this arm and user n . Also it is obvious that $\tau_0 \geq \tau_{b,n}$ based on the selection of $\tau_{b,n}$. We know that for each sample we have at most $4C+1$ independent noises based on Feedback-eliciting subalgorithm, so $s \leq pC\tau_0$ where p is a constant number. Also for any sample we have two Laplace noises (in this model Laplace noise applies one step later and the number of Laplace noises has been reduced), so $l \leq 2\tau_0$.

Set $u = \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b,n}}}$ and $v = \sqrt{\frac{4 \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_{n,b}}}$ and we know that

$$\Pr[|\bar{w}| \geq u] + \Pr[|\bar{z}| \geq v] \geq \Pr[|\bar{w} + \bar{z}| \geq u + v]$$

using union bound inequality. As u, s are the same as u, s in the proof of theorem 1, we can follow the same steps and say

$$\Pr[|\bar{w}| \geq u] \leq \delta / KNB.$$

For $\bar{z}$ we have that

$$\Pr[|\bar{z}| \geq v] \leq 2 \exp(-\frac{v^2 \epsilon^2 \tau_0^2}{2l}) \quad (14)$$

$$\begin{aligned} &= 2 \exp(-(\frac{4 \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_{n,b}}) \frac{\epsilon^2 \tau_0^2}{2l}) \\ &\leq 2 \exp(-(\frac{4 \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_0}) \frac{\epsilon^2 \tau_0^2}{4\tau_0}) \quad (15) \\ &= 2 \exp(-\log(\frac{2KNB}{\delta})) \\ &= \frac{\delta}{KNB} \end{aligned}$$

where the inequality 14 is a direct result of lemma 9 if $v \leq \frac{1}{\epsilon \tau_0}$, which is still true because based on the grid and size of the first batch we have that $\tau_{b,n}$ is at least $O(\sqrt{T'}/K)$, and $T' \gg \log(KNB/\delta)$, K , so $v \leq \frac{1}{\epsilon} \leq \frac{1}{\epsilon \tau_0}$ ($l \geq \tau_0$). Also Inequality 15 holds due to the fact that $\tau_{b,n} \leq \tau_0$ and $l \leq 2\tau_0$. Now, we proved that

$$\begin{aligned} \Pr[|\mu_{i,n} - \hat{\mu}_{i,n,b}| \leq \gamma_{\tau_{b,n}}] &= \Pr[|\bar{w} + \bar{z}| \leq u + v] \\ &\leq \Pr[|\bar{w}| \geq u] + \Pr[|\bar{z}| \geq v] \\ &\leq \frac{2\delta}{KNB} \end{aligned}$$

Now by using a union bound on all arms $i \in [K]$, all users $n \in [N]$ and all batches $b \in [B]$ the theorem is proven. $\square$

Now, we have all the necessary tools to prove theorem 3, like how we proved 1 and we omit the repeated parts.

Proof of Theorem 3.

PROOF. Similar to the proof of Theorem 1, assuming $|\mu_{i,n} - \hat{\mu}_{i,n,b}| \geq \gamma_{\tau_b}$ for all i, n, b , based on 3 in each action during this batch we have the regret of at most $4\gamma_{\tau_{b-1}}$. Now

$$\begin{aligned} \gamma_{\tau_{b-1}} &= \sqrt{\frac{2pC \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} + \sqrt{\frac{4 \log(\frac{2KNB}{\delta})}{\epsilon^2 \tau_{b-1}}} \\ &= (\sqrt{pC} + \frac{\sqrt{2}}{\epsilon}) \sqrt{\frac{2 \log(\frac{2KNB}{\delta})}{\tau_{b-1}}} \\ &\leq (\sqrt{pC} + \frac{\sqrt{2}}{\epsilon}) \sqrt{\frac{2 \log(\frac{2KNB}{\delta})}{t_{b-1}/K - b}} \quad (16) \\ &\approx (\sqrt{pC} + \frac{\sqrt{2}}{\epsilon}) \sqrt{\frac{2 \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} \end{aligned}$$

where inequality 16 holds based on lemma 1. Now using this bound and the fact that in this batch we select $\alpha(4C+2)D_b$ arms each with a sub-optimality of at most $4\gamma_{\tau_{b-1}}$, we bound the regret for b -th batch

$$\begin{aligned} R_b &= 4\alpha(4C+2)D_b\gamma_{\tau_{b-1}} \\ &\leq 4\alpha pC t_b (\sqrt{pC} + \frac{\sqrt{2}}{\epsilon}) \sqrt{\frac{2 \log(\frac{2KNB}{\delta})}{t_{b-1}/K}} \quad (17) \\ &= 4\alpha pCa (\sqrt{pC} + \frac{\sqrt{2}}{\epsilon}) \sqrt{2K \log(\frac{2KNB}{\delta})} \end{aligned}$$

where inequality 17 holds because $t_b \geq D_b$ and $pC \geq (4C+2)$. As $T' = T/\alpha(4C+2) \leq T/\alpha C$ and $a = \Theta(T'^{\frac{1}{2-2^{1-M}}})$ we have

$$\begin{aligned} R_b &= O(\alpha C \sqrt{T/\alpha C} (\sqrt{C} + \frac{1}{\epsilon}) \sqrt{K \log(\frac{2KNB}{\delta})}) \\ &= O((\sqrt{C} + \frac{1}{\epsilon}) \sqrt{\alpha C K T \log(\frac{2KNB}{\delta})}) \end{aligned}$$

for $b > 1$. In the first batch, we suffer a regret of at most 1 for each action, so

$$\begin{aligned} R_1 &= O(4\alpha pCa) \\ &= O(4\alpha pC \sqrt{T/\alpha C}) \\ &= O(\sqrt{T\alpha C}). \\ &\leq R_b \quad (18) \end{aligned}$$

Now we can say that

$$\sum_{b=1}^B R_b = O(B(\sqrt{C} + \frac{1}{\epsilon}) \sqrt{\alpha C K T \log(\frac{2KNB}{\delta})})$$

and the total regret of all instances will be

$$O((1 - \frac{2}{T})NB(\sqrt{C} + \frac{1}{\epsilon}) \sqrt{\alpha C K T \log(\frac{2KNB}{\delta})} + \frac{2}{T}NT).$$

As $B = \log \log T'$ we have proven the theorem. $\square$